TUTORIAL 5: KINEMATICS OF A PARTICLE

ME 1060: DYNAMICS

Conservation of Linear Momentum for a system of particles

When the sum of the external impulses acting on a system of particles is zero, Eq. 15-6 reduces to a simplified form, namely,

$$\sum m_i(v_i)_1 = \sum m_i(v_i)_2$$

This equation is referred to as the *conservation of linear momentum*. It states that the total linear momentum for a system of particles remains constant during the time period t_1 to t_2 . Substituting

$mv_G = \sum m_i v_i$ into,

We can also write

$(v_G)_1 = (v_G)_2$

which indicates that the velocity v_G of the mass center for the system of particles does not change if no external impulses are applied to the system.

The conservation of linear momentum is often applied when particles collide or interact. For application, a careful study of the free-body diagram for the entire system of particles should be made in order to identify the forces which create either external or internal impulses and thereby determine in what direction(s) linear momentum is conserved.

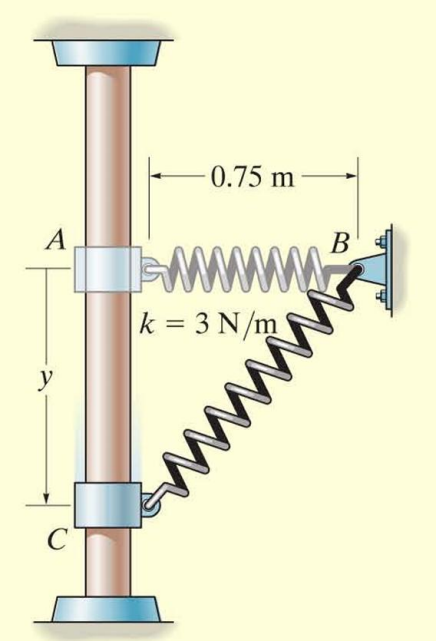
As stated earlier, the internal impulses for the system will always cancel out, since they occur in equal but opposite collinear pairs. If the time period over which the motion is studied is very short, some of the external impulses may also be neglected or considered approximately equal to zero. The forces causing these negligible impulses are called non-impulsive forces. By comparison, forces which are very large and act for a very short period of time produce a significant change in momentum and are called impulsive forces. They, of course, cannot be neglected in the impulse-momentum analysis. Impulsive forces normally occur due to an explosion or the striking of one body against another, whereas non-impulsive forces may include the weight of a body, the force imparted by a slightly deformed spring having a relatively small stiffness, or for that matter, any force that is very small compared to other larger (impulsive) forces. When making this distinction between impulsive and non-impulsive forces, it is important to realize that this only applies during the time t_1 to t_2 ' To illustrate, consider the effect of striking a tennis ball with a racket as shown in the photo. During the *very short* time of interaction, the force of the racket on the ball is impulsive since it changes the ball's momentum drastically. By comparison, the ball's weight will have a negligible effect on the

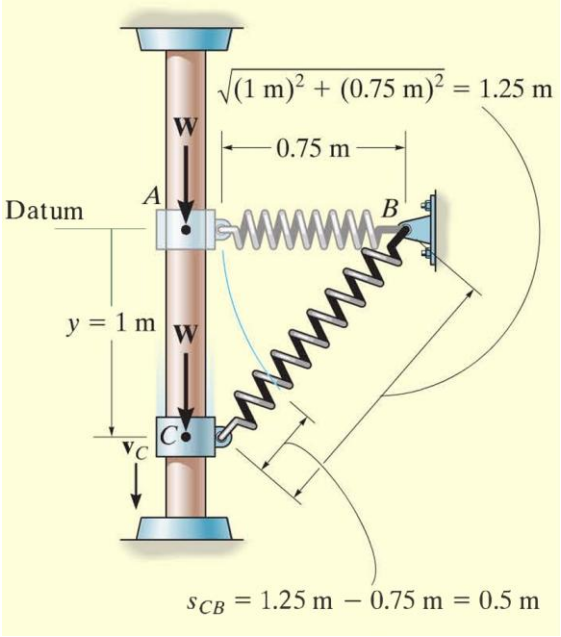
Question 1

A smooth 2-kg collar, shown in figure, fits loosely on the vertical shaft. If the spring is unstretched when the collar is in the position A, determine the speed at which the collar is moving when y = 1m, if

(a) it is released from rest at A, and

(b) it is released at A with an upward velocity v_A = 2 m/s.





SOLUTION

Part (a) Potential Energy. For convenience, the datum is established through AB, Fig. 14–23*b*. When the collar is at C, the gravitational potential energy is $-(mg)y$, since the collar is *below* the datum, and the elastic potential energy is $\frac{1}{2}ks_{CB}^2$. Here $s_{CB} = 0.5$ m, which represents the *stretch* in the spring as shown in the figure.

Conservation of Energy.

$$T_A + V_A = T_C + V_C$$
$$0 + 0 = \tfrac{1}{2}mv_C^2 + \left\{\tfrac{1}{2}ks_{CB}^2 - mgy\right\}$$
$$0 + 0 = \left\{\tfrac{1}{2}(2\text{ kg})v_C^2\right\} + \left\{\tfrac{1}{2}(3\text{ N/m})(0.5\text{ m})^2 - 2(9.81)\text{ N}(1\text{ m})\right\}$$
$$v_C = 4.39\text{ m/s}\downarrow \qquad \textit{Ans.}$$

This problem can also be solved by using the equation of motion or the principle of work and energy. Note that for *both* of these methods the variation of the magnitude and direction of the spring force must be taken into account (see Example 13.4). Here, however, the above solution is clearly advantageous since the calculations depend *only* on data calculated at the initial and final points of the path.

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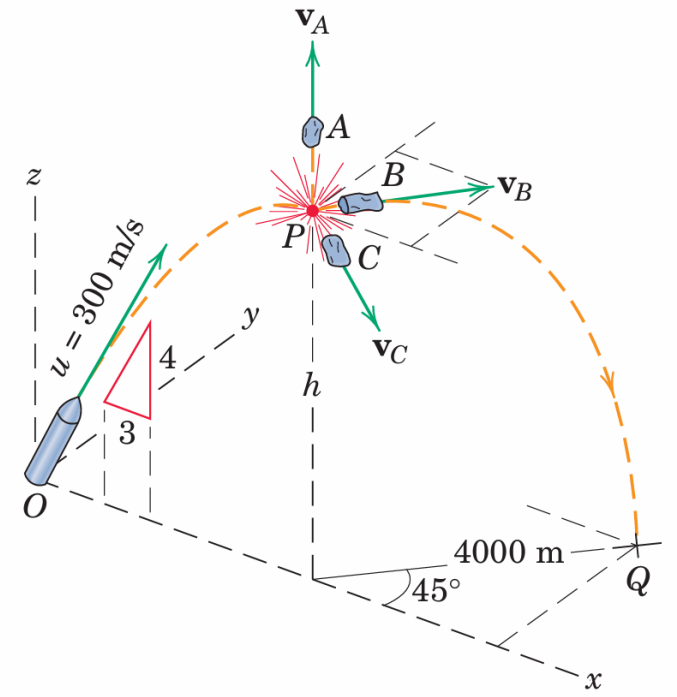
Part (b) Conservation of Energy. If $v_A = 2$ m/s, using the data in Fig. 14–23*b*, we have

$$T_A + V_A = T_C + V_C$$
$$\tfrac{1}{2}mv_A^2 + 0 = \tfrac{1}{2}mv_C^2 + \left\{\tfrac{1}{2}ks_{CB}^2 - mgy\right\}$$
$$\tfrac{1}{2}(2\text{ kg})(2\text{ m/s})^2 + 0 = \tfrac{1}{2}(2\text{ kg})v_C^2 + \left\{\tfrac{1}{2}(3\text{ N/m})(0.5\text{ m})^2 - 2(9.81)\text{ N}(1\text{ m})\right\}$$
$$v_C = 4.82\text{ m/s}\downarrow \qquad \textit{Ans.}$$

NOTE: The kinetic energy of the collar depends only on the *magnitude* of velocity, and therefore it is immaterial if the collar is moving up or down at 2 m/s when released at A.

Question 2

A shell with a mass of 20 kg is fired from point O, with a velocity $u = 300\ m/s$ in the vertical $x - z$ plane at the inclination shown. When it reaches the top of its trajectory at P, it explodes into three fragments A, B, and C. Immedi ately after the explosion, fragment A is observed to rise vertically a distance of $500\ m$ above P, and fragment B is seen to have a horizontal velocity v_B and eventually lands at point Q. When recovered, the masses of the fragments A, B, and C are found to be 5, 9, and 6 kg, respectively. Calculate the velocity which fragment C has immediately after the explosion. Neglect atmospheric resistance.



Solution

From our knowledge of projectile motion, the time required for the shell to reach P and its vertical rise are

$$t = \frac{u_z}{g}\frac{300\left(\frac{4}{5}\right)}{9.81} = 24.5$$

$$h = \frac{u_z^2}{2g} = \frac{\left[(300)\left(\frac{4}{5}\right)\right]^2}{2(9.81)} = 2940\ m$$

The velocity of A has the magnitude

$$v_A = \sqrt{2gh_A} = \sqrt{2(9.81)(500)} = 99.0\ m/s$$

With no z-component of velocity initially, fragment B requires 24.5 s to return to the ground. Thus, its horizontal velocity, which remains constant, is

$$v_B = \frac{s}{t} = \frac{4000}{24.5} = 163.5\ m/s$$

Since the force of the explosion is internal to the system of the shell and its three fragments, the linear momentum of the system remains unchanged during the explosion. Thus,

$$mv = m_A v_A + m_B v_B + m_C v_C$$

$$20(300)\left(\frac{3}{5}\right)i = 5(99.0k) + 9(163.5)(i\cos 45 + j\sin 45^o) + 6v_C$$

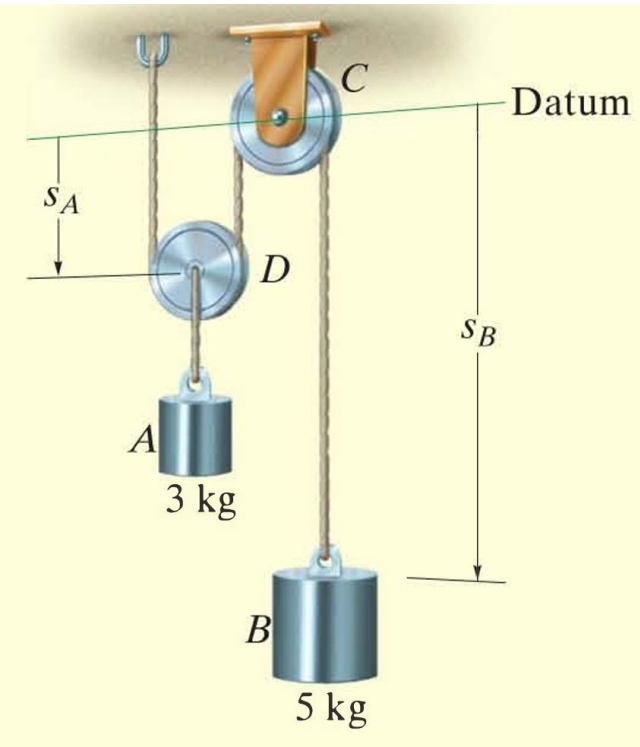
$$6v_c = 2560\,i - 1040\,j - 495k$$

$$v_c = 427\,i - 173.4\,j - 82.5\,k\ \ m/s$$

$$v_c = \sqrt{(427)^2 + (173.4)^2 + (82.5)^2} = 468\ \ m/s$$

Question 3

Blocks A and B shown in Figure have a mass of 3 kg and 5 kg, respectively. If the system is released from rest, determine the velocity of block B in 6 s. Neglect the mass of the pulleys and cord.



Free-Body Diagram.

Since the weight of each block is constant, the cord tensions will also be constant. Furthermore, since the mass of pulley D is neglected, the cord tension $T_A = 2T_B$. Note that the blocks are both assumed to be moving downward in the positive coordinate directions, s_A and s_B.

Principle of Impulse and Momentum.

Block *A*:

$$(+\downarrow) \qquad m(v_A)_1 + \Sigma \int_{t_1}^{t_2} F_y\, dt = m(v_A)_2$$

$$0 - 2T_B(6\text{ s}) + 3(9.81)\text{ N}(6\text{ s}) = (3\text{ kg})(v_A)_2 \qquad (1)$$

Block *B*:

$$(+\downarrow) \qquad m(v_B)_1 + \Sigma \int_{t_1}^{t_2} F_y\, dt = m(v_B)_2$$

$$0 + 5(9.81)\text{ N}(6\text{ s}) - T_B(6\text{ s}) = (5\text{ kg})(v_B)_2 \qquad (2)$$

Kinematics. Since the blocks are subjected to dependent motion, the velocity of A can be related to that of B by using the kinematic analysis discussed in Sec. 12.9. A horizontal datum is established through the fixed point at C, Fig. 15–6*a*, and the position coordinates, s_A and s_B, are related to the constant total length l of the vertical segments of the cord by the equation

$$2s_A + s_B = l$$

Taking the time derivative yields

$$2v_A = -v_B \qquad (3)$$

As indicated by the negative sign, when B moves downward A moves upward. Substituting this result into Eq. 1 and solving Eqs. 1 and 2 yields

$$(v_B)_2 = 35.8\text{ m/s} \downarrow \qquad \textit{Ans.}$$

$$T_B = 19.2\text{ N}$$

NOTE: Realize that the *positive* (downward) direction for $\mathbf{v}_A$ and $\mathbf{v}_B$ is *consistent* in Figs. 15–6*a* and 15–6*b* and in Eqs. 1 to 3. This is important since we are seeking a simultaneous solution of equations.

